Metrics Match Consequences

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable metric-council decision record.

Trace threshold counts,apply costs and explain what the metrics cannot establish.

Keep one habit: W1–W3/W5 → selected fresh four-score task → project metric suite → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Train, Validation, Test

> **My remembered idea, example and correction**

Train a Regression Model

> **My remembered idea, example and correction**

Regression and Residuals

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A0.9 blocked, B0.7 clear, C0.6 blocked, D0.4 blocked, E0.3 clear, F0.1 clear; positive when score>=threshold. At0.5:TP 2/FP 1/FN 1/TN 2, 3 alerts; at0.3:TP 3/FP 2/FN 0/TN 1, 5 alerts. Both4/6 correct. Main FP 1/FN 4 costs5 and2; changed FP 4/FN 1 costs5 and8. Separate picture 4 scores0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy; threshold 0.7/0.4 under FP 1/FN 5 cost5/1, swapped FP 5/FN 1 cost1/5.

  Council capacity 3is a separate stated batch constraint; the independent four-score task has no supplied capacity.

- Which IDs are positive at .5 and .3?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A0.9 blocked, B0.7 clear, C0.6 blocked, D0.4 blocked, E0.3 clear, F0.1 clear; positive when score>=threshold. At0.5:TP 2/FP 1/FN 1/TN 2, 3 alerts; at0.3:TP 3/FP 2/FN 0/TN 1, 5 alerts. Both4/6 correct. Main FP 1/FN 4 costs5 and2; changed FP 4/FN 1 costs5 and8. Separate picture 4 scores0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy; threshold 0.7/0.4 under FP 1/FN 5 cost5/1, swapped FP 5/FN 1 cost1/5.

  Council capacity 3is a separate stated batch constraint; the independent four-score task has no supplied capacity.

- Give TP, FP, FN, TN at both **threshold**s.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A0.9 blocked, B0.7 clear, C0.6 blocked, D0.4 blocked, E0.3 clear, F0.1 clear; positive when score>=threshold. At0.5:TP 2/FP 1/FN 1/TN 2, 3 alerts; at0.3:TP 3/FP 2/FN 0/TN 1, 5 alerts. Both4/6 correct. Main FP 1/FN 4 costs5 and2; changed FP 4/FN 1 costs5 and8. Separate picture 4 scores0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy; threshold 0.7/0.4 under FP 1/FN 5 cost5/1, swapped FP 5/FN 1 cost1/5.

  Council capacity 3is a separate stated batch constraint; the independent four-score task has no supplied capacity.

- Calculate costs with FP cost 1 and FN cost 4.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A0.9 blocked, B0.7 clear, C0.6 blocked, D0.4 blocked, E0.3 clear, F0.1 clear; positive when score>=threshold. At0.5:TP 2/FP 1/FN 1/TN 2, 3 alerts; at0.3:TP 3/FP 2/FN 0/TN 1, 5 alerts. Both4/6 correct. Main FP 1/FN 4 costs5 and2; changed FP 4/FN 1 costs5 and8. Separate picture 4 scores0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy; threshold 0.7/0.4 under FP 1/FN 5 cost5/1, swapped FP 5/FN 1 cost1/5.

  Council capacity 3is a separate stated batch constraint; the independent four-score task has no supplied capacity.

- Why not select the **threshold** with the lowest final-test cost?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- For this new four-record task, "same cost"means the main rule:false alarm 1 point, miss 4 points. Use blocked positive and score>=threshold, including equality at0.5 and0.2. No review-capacity constraint is supplied for this task; do not import the separate council batch capacity. Report the preference among these two thresholds under the stated fictional costs without a final-test or real-world claim.

- **Score**s/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Approve the Plant Classifier **metric suite**: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Count the **consequence**s for **Threshold**s and **consequence**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Threshold**s and **consequence**s. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.
- Connect metrics to **consequence**s.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Missing a diseased plant is considered more costly than a **false alarm**, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 25 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Cost and decision rules: blocked is positive; predict blocked when score is at least the threshold, including equality. A false positive costs 1 unit and a false negative costs 4 units. Total cost = FP + 4 × FN. Compare both thresholds on exactly these four records.

- **Score**s/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Translate a **threshold** choice into missed-leaf and false-alarm costs.
- **Score**s/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/25

Metrics Match Consequences

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Threshold0.7

TP 1, FN 1, FP 0, TN 2: cost5.

2. Threshold0.4

TP 2, FN 0, FP 1, TN 1: cost1.

3. Choose for stated costs

Threshold0.4 has smaller validation cost on these four cases.

CHANGE ONE THING

Miss and false-alarm costs swap to1 and 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Translate a **threshold** choice into missed-leaf and false-alarm costs.
- Which labelled step matches this detail? **Threshold**0.4 has smaller validation cost on these four cases.
- 1. **Threshold**0.4
- 2. Choose for stated costs
- 3. **Threshold**0.7
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Choose for stated costs **Threshold**0.4 has smaller validation cost on these four cases. Read the picture in order.

Threshold0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Miss and false-alarm costs swap to1 and 5. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches. Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Repair and keep the record

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the relevant picture instead of repeating every task.

01. Metrics Match Consequences: the case

AI ACADEMY / AI-4/25 · 01 · LOOK AT THE FACTS

Metrics Match Consequences: the case

The situation

Translate a threshold choice into missed-leaf and false-alarm costs.

Supplied evidence

Scores/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at score>=threshold. Miss cost5; false-alarm cost1.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain metrics match consequences using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

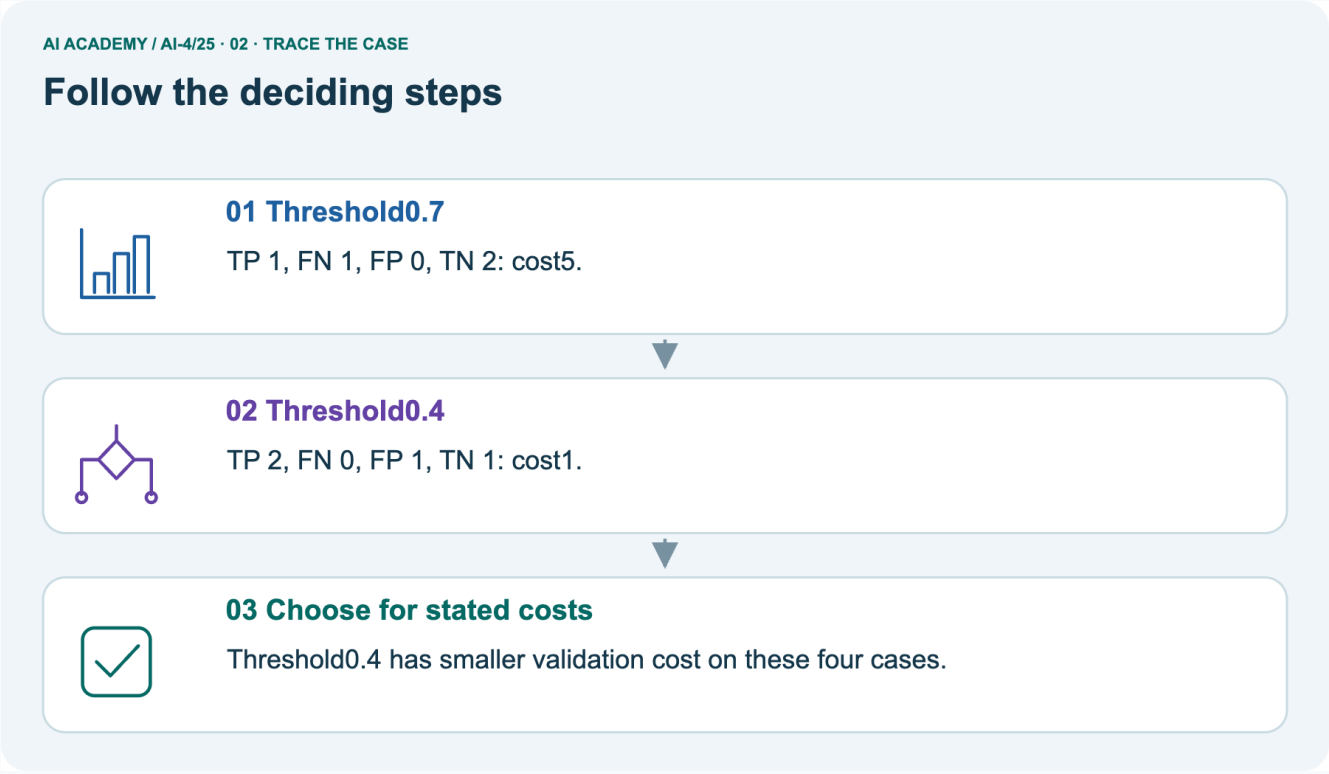
Translate a **threshold** choice into missed-leaf and false-alarm costs. Supplied facts: **Score**s/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Threshold0.7: TP 1, FN 1, FP 0, TN 2: cost5. → **Threshold**0.4: TP 2, FN 0, FP 1, TN 1: cost1. → Choose for stated costs: **Threshold**0.4 has smaller validation cost on these four cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Thresholds and consequences at a glance

AI ACADEMY / AI-4/25 · 03 · INSPECT THE LESSON MODEL

Thresholds and consequences at a glance

Part	Evidence or action	Interpretation
Threshold .5	One false alarm one miss	Four of six correct
Threshold .3	Two false alarms no misses	Four of six correct
Decision rule	Include equality	Record before evaluation

Lowering the threshold here finds D but adds E as a false alarm. The two accuracies are equal while consequences differ. Precision is 2/3 at .5 and 3/5 at .3; recall rises from 2/3 to 1. This behavior follows these fixed scores and records, not a promise of future performance.

Follow the labelled picture with the matching explanation. Open the vector diagram

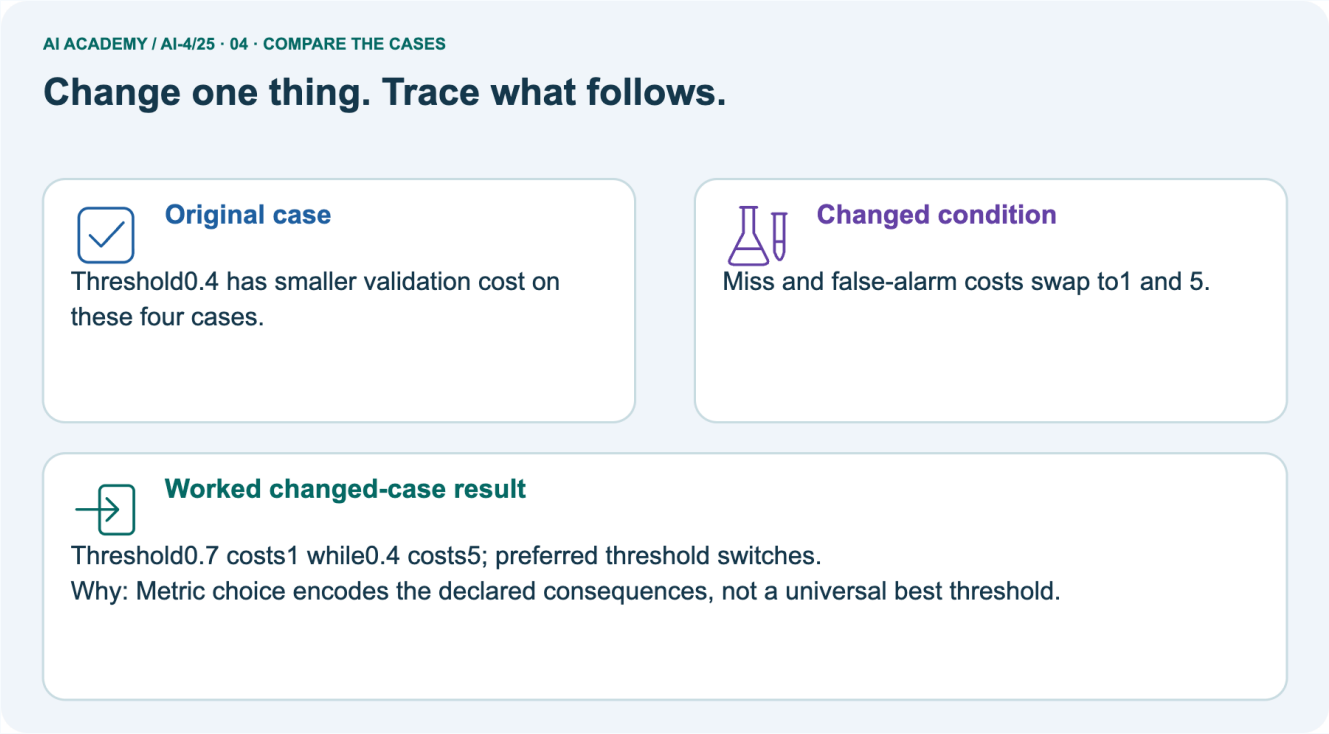
Thresholds and **consequence**s at a glance. **Threshold** .5 | One **false alarm** one miss | Four of six correct. **Threshold** .3 | Two **false alarm**s no misses | Four of six correct.

Decision rule | Include equality | Record before evaluation

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

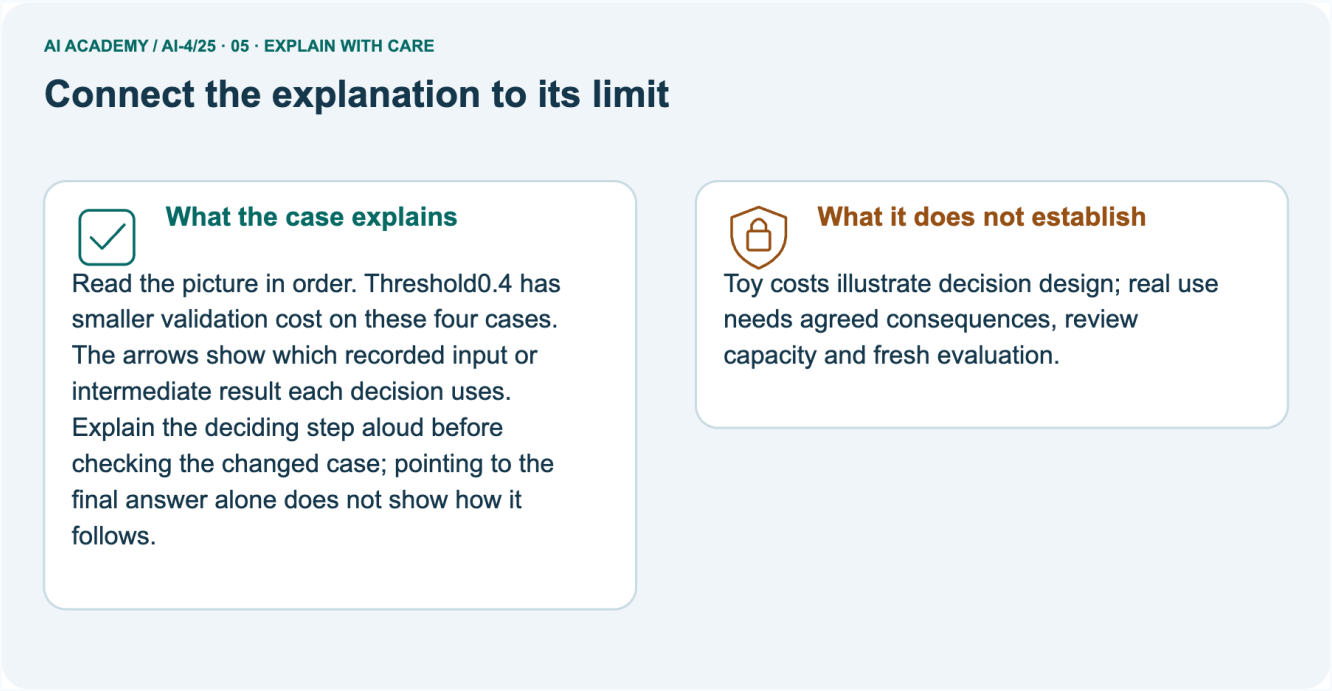
Original case: **Threshold**0.4 has smaller validation cost on these four cases. Changed condition: Miss and false-alarm costs swap to1 and 5. Expected result: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Metrics Match Consequences: the case

02. Follow the deciding steps

03. Thresholds and consequences at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.